

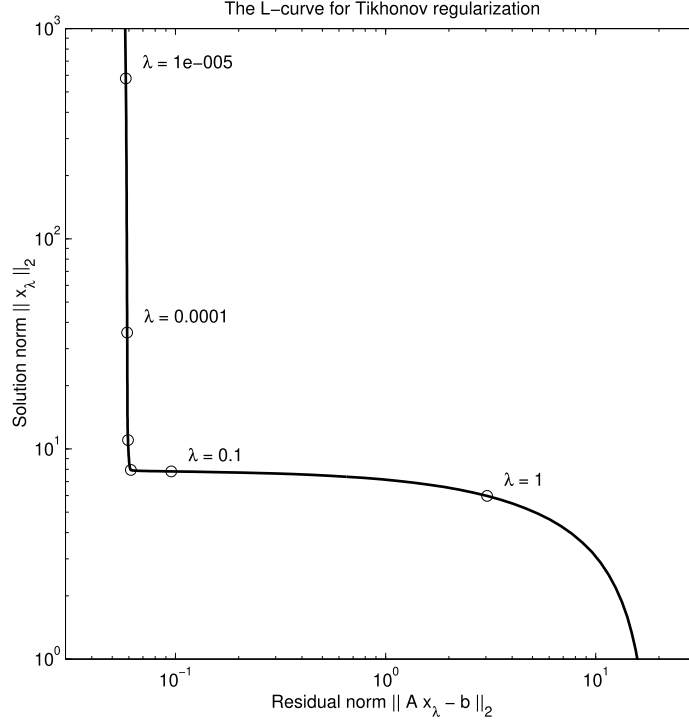


Figure 9.1: Representative L-Curve Plot. A representative L-curve plot shown with various regularization parameter (λ) values is presented. The optimal value of the regularization parameter is the point corresponding to maximum curvature. This point corresponds to a kink in the L-curve. Generally, this number is a fractional number that is less than 1. In the text, the λ -parameter is equivalently labeled as τ . Plot is taken from Ref. [133].

In practice, the user defines a constant number of iterations to scan between a range of possible τ values. At each iteration, the unfolding is performed with a different τ value, and the corresponding L_x^{curve} and L_y^{curve} values are recorded. The curvature of the L-curve is then calculated at each point, and the τ value that maximizes the curvature is selected as the optimal value. The optimal value is then used during the final iteration of the unfolding to obtain the unfolded distribution. This final step completes the unfolding process. The unfolding process is correctly verified with a closure test, with the results shown in Appendix D.

9.2 Unfolding Ingredients

For each observable of interest, the following ingredients are required to perform unfolding.

Full Detector-Level Distribution, R The distribution comprises all reconstructed events filled with MC simulated events that pass the detector-level selection.

Full Particle-Level Distribution, T The distribution comprises all particle-level events filled with MC simulated events that pass the particle-level selection.